

A Dual-IMU Haptic Monitor for Post-Stroke Genu Recurvatum Correction

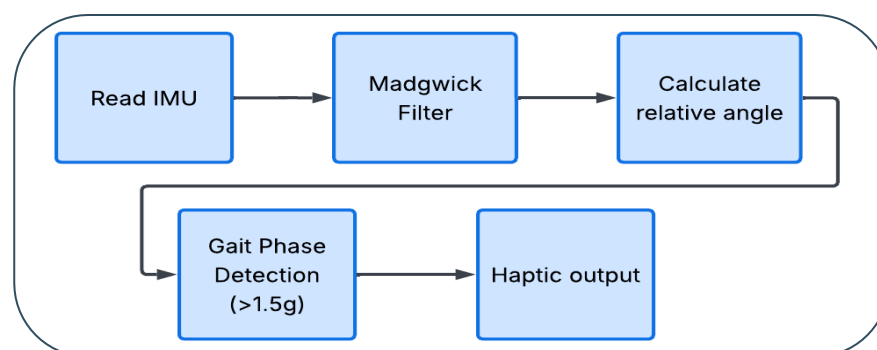
Clinical background

- Target population:** Affects **40–68%** of paretic stroke patients [1]
- Genu Recurvatum:** Knee hyperextension during the gait stance phase
- Consequences:** Chronic pain, tissue damage, inefficient asymmetrical gait [2]
- Root Cause:** Patients cannot "feel" the 0° anatomical limit
- Current standard:** Rigid braces, which risk muscle atrophy [3]



Materials & Methods

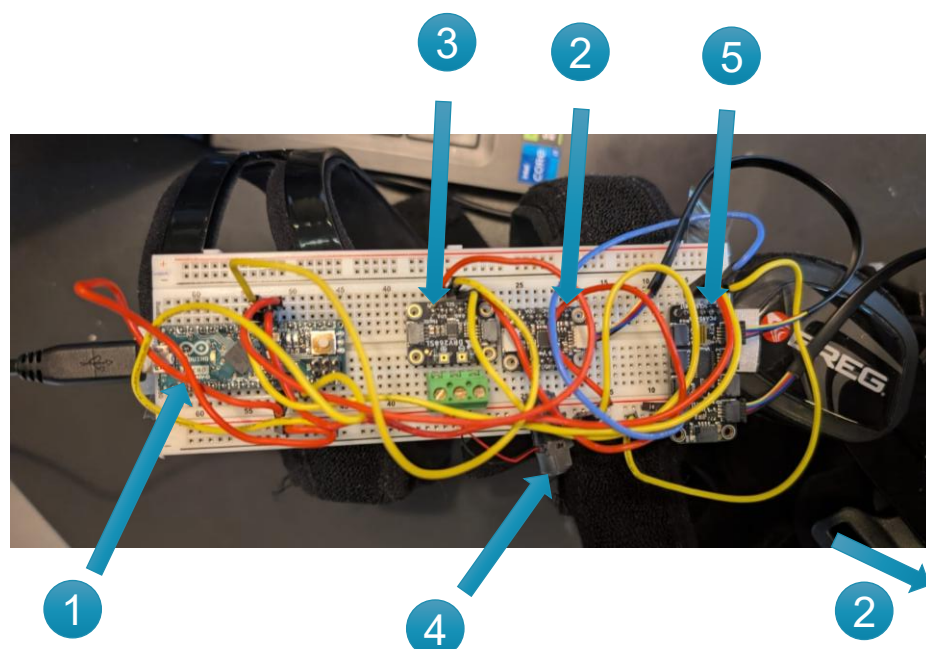
1. Arduino
2. 2x MPU6050 IMU
3. DRV2605L Haptic Driver
4. Drake Haptic Actuator
5. Multiplexer PCA9548A



Digital twin & prototype



Red => Knee hyperextension



Discussion & Conclusion

- Dual-Loop Feedback:** The patient feels a warning vibration that gets stronger as the knee straightens, while the therapist monitors live 3D gait data
- Objective Mapping:** Virtual 3D mirrors kinematics and turns red upon hyperextension.
- Anatomical Calibration:** Sets the initial stance to a 0° baseline.
- Gravity-Based Kinematics:** Projects orientation onto the global gravity vector, bypassing 6-DOF Yaw-drift.

Future work

- 9-DOF Upgrade:** Transition to BNO085 sensors to eliminate Yaw drift
- Wireless Nodes:** Implement Bluetooth Low Energy (BLE) microcontrollers to eliminate wiring
- Clinical validation:** long term testing sessions with patient target group

